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Trends in Biotechnology

Review

# Micro-aeration: an attractive strategy to facilitate anaerobic digestion

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**Micro-aeration can facilitate anaerobic digestion (AD) by regulating microbial communities and promoting the growth of facultative taxa, thereby increasing methane yield and stabilizing the AD process. Additionally, micro-aeration contributes to hydrogen sulfide stripping by oxidization to produce molecular sulfur or sulfuric acid. Although micro-aeration can positively affect AD, it must be strictly regulated to maintain an overall anaerobic environment that permits anaerobic microorganisms to thrive. Even so, obligate anaerobes, especially methanogens, could suffer from oxidative stress during micro-aeration. This review describes the applications of micro-aeration in AD and examines the cutting-edge advances in how methanogens survive under oxygen stress. Moreover, barriers and corresponding solutions are proposed to move micro-aeration technology closer to application at scale.**

## Highlights

Micro-aeration represents a promising strategy to facilitate anaerobic digestion (AD), as it enhances hydrolysis, improves methane yield, oxidizes specific pollutants, and thereby overall improves AD.

Low oxygen solubility in water, rapid consumption of dissolved oxygen by facultative microbes, the formation of microbial aggregates, and the intrinsic tolerance of some methanogenic species to oxygen are suggested as potential mechanisms for unimpaired AD process in response to oxygen exposure.

Fundamental understanding of anaerobic microorganisms' tolerance to oxygen stress and long-term pilot scale tests of effects of micro-aeration on AD are needed in order to move micro-aeration technology closer to applications.

## AD and oxygen

AD is a natural process occurring in sites where oxygen is absent – for example, sediments, and wetlands – or in manmade anaerobic environments such as biogas reactors [1]. AD has been used for bioenergy production since the 10th century, with biogas being used to heat bath water in Assyria [2]. Nowadays, AD technology plays an increasingly important role in organic waste disposal as well as bioenergy production (Box 1) [3]. Due to the energy crisis and global warming, biogas as a **carbon-neutral energy** (see Glossary) carrier presents unprecedented development opportunities [4]. The global power production capacity of biogas plants was about 19.5 GW in 2019 [5] and is expected to reach 29.5 GW in 2022 [6]. However, some challenges in AD, such as low biomethane yield, long fermentation period, and easy acidification, should be addressed to improve the process. In addition, the complex microbial community, comprising various microbial groups (i.e., hydrolytic bacteria, acidogens, acetogens, and **methanogens**) living in symbiotic association, require delicate balancing to maintain the stability of the process [7].

Oxygen strongly inhibits the growth and metabolism of **obligate anaerobic microorganisms** (Box 2), especially methanogens, as they cannot protect themselves from **reactive oxygen species (ROS)** such as superoxide and hydrogen peroxide [10,11]. However, exposure to oxygen is inevitable in the practical operation of the AD reactor, as the reactor operates in an aerobic environment. Furthermore, the beneficial effects of micro-aeration [defined as limited oxygen exposure to the AD system (Box 3)] on the AD process have been reported in numerous studies [12–14]. As reported, micro-aeration in an AD system enhanced the hydrolysis of substrate [12], improved overall methane yield [13], stabilized AD [7], enabled hydrogen sulfide stripping [14], etc. and represents an attractive strategy to improve AD. However, some studies reported no or even negative effects of micro-aeration on AD [15,16]. These conflicting results may be explained by the different substrate and oxygen dosing method used, as well as the unclear underlying mechanisms of how micro-aeration facilitates the AD process.

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Box 1. AD for biogas production

AD is a microbially mediated process during which biodegradable organic matter is converted to a gaseous product (i.e., biogas) via several steps (i.e., hydrolysis, acidogenesis, acetogenesis and methanogenesis) [4]. Various microorganisms are involved, including hydrolytic bacteria that produce hydrolytic enzymes for the initial decomposition of polymeric organics, while others, such as acidogens, acetogens, and methanogens, metabolize the intermediate compounds to produce the final biogas product [1]. Although AD is a promising technology for the mineralization of organic matter and bioenergy production, some challenges (e.g., low biomethane yield, long fermentation period, process stability) exist during the AD process. An attractive way to overcome these challenges is to regulate microbial communities with **microbial resource management** [8]. Microbial resource management can be achieved via two strategies; namely, bioaugmentation (i.e. adding specific adapted microbial cultures) and the application of operational parameters, which the promote establishment of specific microbial communities [9]. Both strategies aim to establish a microbial consortium with a desired composition to achieve specific functions (e.g., avoiding inhibition, increasing biodegradability of organics). The use of microbial resource management to regulate the AD process is now attracting increasing attention and has become a hot topic of research.

Considering the boom in AD technology for organic waste disposal as well as renewable energy production, micro-aeration is an attractive strategy to improve the AD process and has gained increased interest. An increasing number of studies have been conducted to evaluate the effects of micro-aeration on AD as well as to uncover the underlying mechanisms [7,10–16]. However, a systematic study to critically summarize the important roles of micro-aeration in facilitating AD as well as to guide future research is still lacking. Herein we review the applications of, barriers to, and relevant solutions for micro-aeration in AD. In addition, the mechanisms of how methanogens survive against oxygen stress and perspectives on the development of micro-aeration will be discussed to give clarity for the potential application of this technology on a large scale.

## The applications of micro-aeration in AD

Studies on the effects of micro-aeration on AD systems date back to the 1990s, when Kato and colleagues investigated the impacts of oxygen exposure on methanogenesis in granular sludge and found that the methanogens showed a certain tolerance to oxygen [21]. Thereafter, Shen and colleagues studied the impact of dissolved oxygen on the activity of anaerobic granules [22] and the performance of a co-culture of **methanotrophs** (aerobic bacteria) and methanogens (strict anaerobic archaea) under oxygen-limited conditions [23]. This study showed that co-cultivation maintained and even improved the metabolic activities of methanogens under oxygen-limited conditions [23]. Recently, micro-aeration has been widely applied during AD for a variety of purposes including hydrolysis enhancement, stability maintenance, hydrogen sulfide stripping, and pollutant degradation acceleration (Figure 1).

### Using micro-aeration to enhance hydrolysis

Hydrolysis performed by **extracellular hydrolytic enzymes**, secreted by mainly aerobic or facultative and some anaerobic bacteria, is usually considered the rate-limiting step of AD, especially for the recalcitrant substrate (e.g., lignocellulosic matter) [24]. Pretreatment of the substrate using mechanical, thermal, chemical, and biological methods has been widely reported [25]. However, economic and environmental considerations limit the application of these techniques at a large scale. Another choice is to improve the activities or concentrations of the extracellular hydrolytic enzymes (e.g., cellulases, xylanase, lipases). In this respect, micro-aerobic pretreatment of substrate or micro-aeration during AD represents a promising option. As reported, micro-aeration conditions may facilitate the diversity and abundance of hydrolytic bacteria generally [26], which secrete more extracellular hydrolytic enzymes (e.g., cellulases, xylanase, lipases) in turn, and eventually results in improved hydrolysis of complex biomasses such as lignocellulose and lipids [19,27]. Micro-aeration could create a unique environment for both **facultative microbes** and obligate anaerobic microorganisms, thus permitting coexistence, during which the facultative microbes could consume the traces of oxygen and ensure anaerobic conditions for obligate anaerobic microorganisms. Consequently, a diverse hydrolytic and fermentative

## Glossary

**Carbon-neutral energy:** energy that produces no net **greenhouse-gas** emissions or carbon footprint. Bioenergy produced using natural $CO_2$-consuming processes such as photosynthesis is an example.
**Electrolytic aeration:** oxygen produced via electrolysis of water.
**Extracellular hydrolytic enzymes:** enzymes that catalyze the cleavage of a covalent bond using water, which are secreted by microbes to function outside the cells.
**Facultative microbes:** microbes that can survive in both oxygen-rich and oxygen-deficient environments.
**Greenhouse gases:** gases able to absorb IR radiation thus warming the Earth and contributing to the greenhouse effect. Carbon dioxide, methane, and nitrous oxide are typical greenhouse gases.
**Hydrogenotrophic methanogens:** methanogens that use principally $H_2$ as the electron donor to drive the reduction of $CO_2$ for $CH_4$ formation.
**Lower explosive limit (LEL):** the minimal concentration of a particular chemical needed to cause an explosion.
**Methanogens:** unicellular archaeal microorganisms that are obligate anaerobes extremely sensitive to oxygen, which produce methane as their metabolic end-product.
**Methanotrophs:** microorganisms using methane as their sole carbon and energy source.
**Microbial resource management:** management of a microbial community by the direct addition of specific species (bioaugmentation) or by changing the operational parameter to promote the growth or exclusion of individual species, with the aim of optimizing specific bioprocesses.
**Obligate anaerobic microorganisms:** organisms that can live only in environments lacking oxygen.
**Oxygen loading:** the amount of oxygen used.
**Reactive oxygen species (ROS):** oxygen-containing radicals that are capable of independent existence with one or more unpaired electrons. Superoxide and hydroxyl radicals, nitric oxide, singlet oxygen, nitrogen dioxide, and peroxynitrite are typical ROS.
**Sulfate-reducing bacteria (SRBs):** anaerobic microorganisms that can oxidize organic matter and reduce

## Box 2. Toxicity of oxygen to obligate anaerobic microorganisms

The primordial atmosphere of Earth was virtually anoxic, and the microbial organisms that appeared had biochemical mechanisms and metabolic pathways that do not use oxygen as the terminal electron acceptor [11]. Aerobes that use molecular oxygen as the electron acceptor for oxidative phosphorylation to produce ATP and water appeared later with the emergence of oxygen in the atmosphere [11,17]. During biological consumption, molecular oxygen randomly captures electrons to form superoxide anions ($O_2$•$^-$; Equation I), which can then form downstream derivatives such as hydrogen peroxide ($H_2O_2$; Equation II) and hydroxyl radicals (•OH; Equations III and IV) [17].

$$O_2 + e^- \rightarrow O_{2^-} \quad [I]$$

$$2O_{2^-} + 2H^+ \rightarrow H_2O_2 + O_2 \quad [II]$$

$$O_{2^-} + H_2O_2 \rightarrow \bullet OH + OH^- + O_2 \quad [III]$$

$$H_2O_2 + Fe^{2+} \rightarrow Fe^{3+} + \bullet OH + OH^- \quad [IV]$$

Aerobes generally possess antioxidant defense systems to resist and mitigate the adverse effects of excessive ROS (i.e., $O_2$•$^-$, $H_2O_2$, and •OH). Conversely, obligate anaerobes usually lack such a defense system, and if ROS are not dealt with in a timely fashion they will cause oxidative damage (Figure I) [17]. Obligate anaerobes are therefore incapable of counteracting and surviving oxidative stress [18].

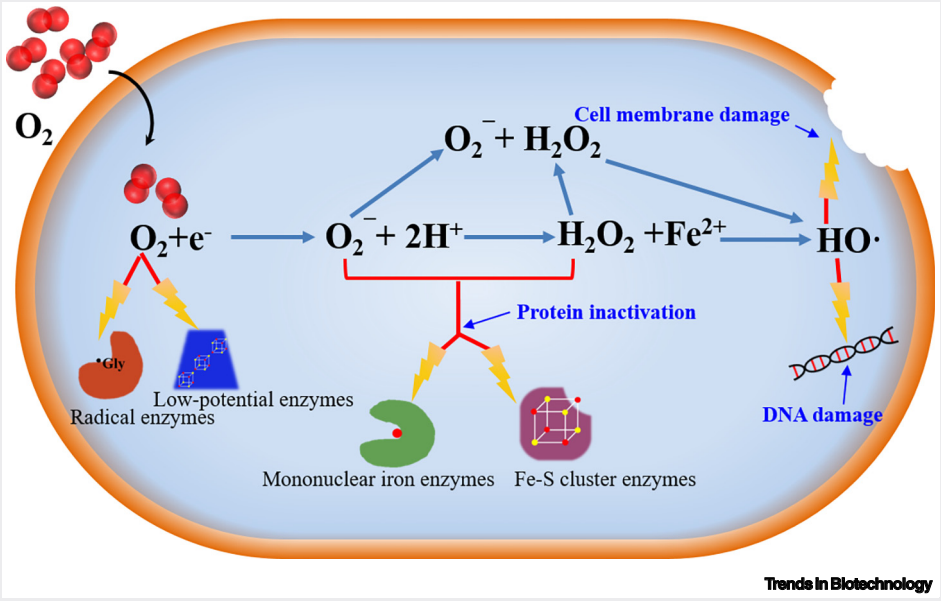



Figure I. Toxicity of oxygen to obligate anaerobic microorganisms. When the microorganisms are exposed to oxygen, molecular oxygen crosses the membrane and accumulates intracellularly. The intracellular oxygen then captures electrons to form superoxide anions and downstream derivatives (e.g., hydrogen peroxide, hydroxyl radicals). These reactive oxygen species (ROS) lead to oxidative damage to microbial activity (or central metabolism) and important cellular molecules (e.g., DNA damage, protein inactivation). Adapted from [18].

sulfate to sulfide, playing a crucial role in metal sulfide precipitation.

**Sulfur-oxidizing bacteria (SOBs):** microorganisms that can oxidize hydrogen sulfide, elemental sulfur, and thiosulfates to produce sulfuric acid.

**Upper explosive limit (UEL):** the maximum concentration of a particular chemical that can cause an explosion.

**Volatile fatty acids (VFAs):** short-chain aliphatic monocarboxylate compounds comprising mainly C2–C6 carboxylic acids, such as acetic acid, propionic acid, and butyric acid.

microbial community will be established. For instance, the phyla Fimicutes and Bacteroidetes are considered the functional consortia for the hydrolysis of hemicellulose, cellulose, and other polysaccharides in lignocellulose [28] as well as excess the sludge of domestic wastewater treatment [29]. It has been reported that the relative abundances of these hydrolytic bacteria increased under micro-aeration [26,30] and they contributed greater hydrolysis of the polymeric

Box 3. Micro-aeration

Micro-aeration in an AD system refers to the introduction of limited amounts of oxygen or air into the system, either during pretreatment of the feedstock or directly into the AD reactor. The introduction of oxygen influences the AD process, either by pretreatment of the feedstock or by improvement of the AD process by altering the composition of the microbial community. The oxygen/air loading in different studies is diverse depending on their aims [10]. If micro-aeration was used during pretreatment, the oxygen/air dosing needs to be relatively high. Conversely, if micro-aeration was applied for hydrolysis enhancement or methane yield improvement, the oxygen/air dosing should be relatively low to avoid serious obstruction of methanogenesis. When hydrogen sulfide stripping was targeted, the oxygen/air dosing should be applied based on the sulfur content in the feedstock. In addition to the oxygen/air dosing, micro-aeration units are also quite diverse. Various units (e.g., $O_2/L_R$ per day, ml/g VS, mg/l, μM) are widely used to describe micro-aeration. A standard method to describe micro-aeration is still lacking. The oxygen level during micro-aeration is usually too low to be detected using an instrument that measures dissolved oxygen, because it is quickly consumed by facultative microorganisms. Instead, oxidation-reduction potential (ORP), which varies linearly with the logarithm of oxygen concentration, could reflect the tiny changes in oxygen concentration in the liquid phase (Figure I) [19]. ORP as a feedback parameter has been used in several studies to precisely control oxygen dosing [7,20]. However, the ORP of the AD system is susceptible to running parameters (e.g., pH, temperature, ionic strength). Therefore, an automatic monitoring and control system should be developed to autocompensate these parameters.

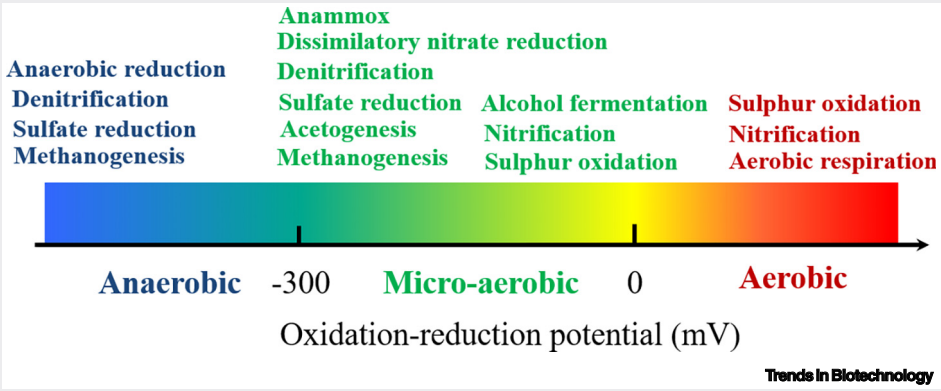



Figure I. Biochemical reactions under different oxidation-reduction potentials (ORPs). ORP lower than –300 mV enables anaerobic conditions, and below –400 mV strict anaerobic conditions occur where strict anaerobes such as methanogens thrive. When the ORP increases to higher than –300 mV, sulfate reduction conditions prevail (when sulfate is present), while at higher than –200 mV denitrification dominates (when nitrate is present) or facultative fermentation occurs. ORP higher than 0 mV indicates aerobic conditions.

compounds to simple sugars followed by acidogenesis of the sugars to produce **volatile fatty acids (VFAs)**. Many acidogenic bacteria also possess hydrolytic ability, secreting extracellular enzymes for decomposition of the complex polymeric compounds (e.g., cellulose, polysaccharide, lipid) to simpler monomers or dimers, which can subsequently be assimilated by these bacteria. Micro-aeration affects not only the hydrolysis of polymeric compounds but also the subsequent fermentation of the monomers, resulting in differing products [10]. VFAs are key products of the acidogenic process. These VFAs are further converted by another group of strict anaerobes – the acetogens – to simpler compounds; namely, acetate and $H_2$-$CO_2$, which are the final methanogenic substrates resulting in the production of biogas [30]. As Lim and colleagues [31] reported, micro-aeration led to increased VFA (especially propionate) concentrations during anaerobic co-digestion of brown water and food waste. Similarly, Xu and colleagues obtained a threefold increase in acetate and butyrate concentration under micro-aeration conditions [32].

## Using micro-aeration to stabilize the AD process

AD is a microbiological process performed by various microorganisms (i.e., hydrolytic bacteria, acidogens, acetogens, methanogens) [33]. During micro-aeration, the microorganisms of AD

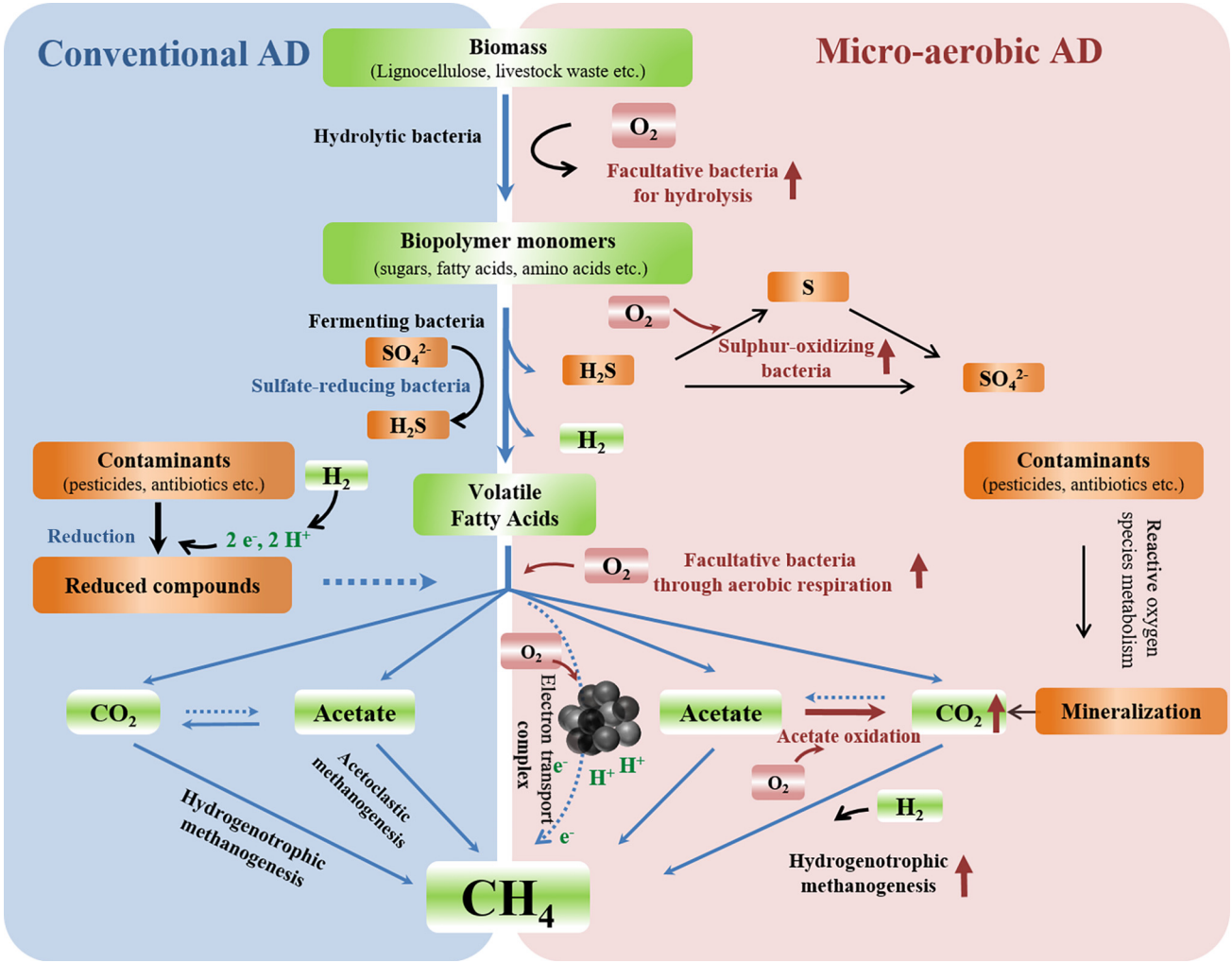



Figure 1. Effects of micro-aeration on anaerobic digestion (AD). In a typical AD system, the complex organic polymers are firstly hydrolyzed into smaller units (e.g., monomers) by the enzymatic activity of enzymes produced by hydrolytic bacteria. Then, the monomers are metabolized into volatile fatty acids (VFAs) and $H_2$ by fermenting acidogenic bacteria. When sulfate is present, $H_2S$ is generated catalyzed by sulfate-reducing bacteria. VFAs then are degraded into acetate, $CO_2$, and $H_2$ via acetogens. Finally, the acetate, $CO_2$, and $H_2$ are used by methanogens for $CH_4$ production via methanogenesis. Micro-aeration could affect the AD process via regulation of microbial communities. The beneficial effects of micro-aeration on AD include hydrolysis enhancement, stripping of hydrogen sulfide, stabilization of the AD process, improved methane yield, and contaminant degradation.

are screened and domesticated to form a unique microbial community with enhanced diversity and activity of obligate and facultative anaerobes, which contribute to the stable running of the AD process. In addition to the enhanced diversity of the microbial community, micro-aeration was reported to maintain stable pH by partial oxidation of VFAs through aerobic respiration [7] or selective oxidation of acetate through acetate-oxidizing bacteria [30] or promotion of the syntrophic relationship between facultative bacteria and **hydrogenotrophic methanogens** [34]. The role of micro-aeration in promoting or reducing VFA accumulation could be strongly associated with **oxygen loading** and organic loading rate (OLR). For example, *Rummeliibacillus* that can secrete multiple cellulases and hemicellulases was found as a dominant microbe in the acidification of lignocellulose under micro-aeration [12]. In addition, *Proteiniphilum* sp. was

discovered as an important taxon in the conversion of excess VFAs into $CO_2$ via aerobic respiration during intermittent micro-aeration in high-OLR AD processes [7,35]. The relative abundance of hydrogenotrophic methanogens (e.g., *Methanobacterium*, *Methanocorpusculum*, *Methanobrevibacter*, and *Methanosarcina*) was also significantly increased under micro-aerobic conditions, promoting the hydrogenotrophic methanogenic pathway, which maintained low hydrogen partial pressure [36]. It was also reported that compounds able to act as electron shuttles (e.g., humic acids) were enhanced and facilitated direct interspecific electron transfer (DIET) and syntrophic collaboration between acetogens and methanogens, which resulted in overall improved AD [30].

### Using micro-aeration to strip hydrogen sulfide

During the AD of sulfur-rich biomass, hydrogen sulfide will be generated via dissimilatory sulfate reduction [37]. Since **sulfate-reducing bacteria (SRBs)** such as the genus *Desulfovibrio* [38] and *Desulfobacter* sp. [39] compete with methanogens on the substrate (mainly acetate and $CO_2$), this results in low methane yield and accumulated hydrogen sulfide [19,40]. In addition, the $H_2S$ in biogas leads to corrosion of reactor equipment and is especially harmful to gas motors where biogas is combusted to produce electricity. Therefore, it is crucial to remove the $H_2S$ before biogas is injected into the gas motor. Conventional $H_2S$ removal for biogas purification is mainly based on physical–chemical methods (e.g., alkaline wash, iron filters containing iron oxide in the form of iron sponges) [41]. An alternative desulfurization method is based on a biological method using **sulfur-oxidizing bacteria (SOBs)** such as *Thiobacillus*, *Halothiobacillus*, *Sulfuricurvum*, and *Acinetobacter* assigned to the phylum Proteobacteria, which oxidize $H_2S$ to $H_2SO_4$ or molecular S. The biological method using SOBs can achieve higher removal efficiency and lower costs without requiring additional chemicals. Usually, biological biogas cleaning is performed in dedicated desulfurization units. Direct desulfurization in biogas reactors, either by the injection of air in the headspace of the reactor or micro-aeration where air is injected directly into the liquid phase of the reactor, has recently drawn attention due to its convenience [37]. Previous studies have shown that the diversity and activity of facultative bacteria associated with sulfide oxidation were enhanced under micro-aeration [37]. Meanwhile, the reaction site of sulfide oxidation was on the wall of the reactor headspace or during the gas–liquid interphase. Thus, *in situ* $H_2S$ removal in AD could be achieved via micro-aeration [37,42,43] and has widely been performed for biomasses with a high sulfide content, such as sewage sludge and livestock waste, realizing significant improvements on hydrogen sulfide elimination. For example, complete biogas desulfurization was achieved in full-scale thermophilic digesters with an air blower in the headspace for sewage sludge treatment after 81 weeks of running under various residue oxygen content (0.2–2.0%) [44], and $H_2S$ removal efficiency above 90% was obtained with air addition in an up-flow anaerobic sludge blanket (UASB) reactor treating domestic sewage [45]. It was also reported that micro-aeration enhanced the methane yield of chicken manure and mitigated $H_2S$ formation simultaneously [46,47].

### Using micro-aeration to accelerate pollutant degradation

The AD process can realize the energetic utilization of municipal and agricultural solid waste, which conventionally contains persistent organic pollutants such as pesticides and antibiotics. In addition, the anaerobic method is also widely used in the treatment of wastewater contaminated with petroleum, pharmaceuticals, dyes, supernatants, etc. [48–51]. These pollutants can be transformed with anaerobic treatment; however, it takes quite a long time to achieve complete mineralization under anoxic conditions [52] as degradation of these pollutants occurs preferably under aerobic conditions. Many pollutants, such as petroleum-based alkanes, lack an active functional group to initiate the degradation process. However, the presence of oxygen can initiate this process: for example, oxidizing a methyl group to carboxylic group or alcohol, with anaerobic

decomposition occurring thereafter. Oxygen molecules as strong oxidants play the terminal acceptor role in the exergonic respiratory chain in various abiotic redox reactions [53]. In addition, micro-aeration enables a wide diversity and high abundance of hydrolytic bacteria, resulting in improved hydrolysis of complex organic matter and increased biodegradability [50]. Micro-aeration in AD has also been applied to accelerate the degradation of contaminants. For example, as the raw material of synthetic compounds such as pesticides, pharmaceuticals, and dyes, 2,4-dinitrophenol is often found in wastewater, and it was demonstrated that redox enzymes under micro-aeration could enhance the degradation of 2, 4-dinitrophenol [50]. In addition, degradation of benzene, toluene, ethylbenzene, and xylenes (BTEX) [49] and surfactant were enhanced under micro-aerobic treatment. Moreover, P450s are considered $O_2$- and NADPH-dependent enzymes [54,55] that are involved in the reductive transformation of contaminants under anoxic conditions [56,57]. Under micro-aeration, the P450s are expected to further participate in the oxidation of the contaminants and are not limited to reductive transformation, although relative evidence remains lacking. Thus, further studies to reveal the important role of P450s in accelerating contaminant degradation under micro-aeration should be conducted.

## Micro-aeration in AD: how methanogens survive oxygen stress

During micro-aeration, obligate anaerobic microorganisms would inevitably suffer oxygen stress. The AD process can be divided into four steps: hydrolysis, acidolysis, acetogenesis, and methanogenesis. Among those steps, methanogenesis is the most sensitive because it can be performed by only a few methanogens [58]. The improved methane yields under micro-aeration reported in previous studies indicated the ability of methanogens to function under slightly elevated redox conditions. Analysis of the mechanisms of how methanogens survive oxygen stress is crucial to guide the industrial application of micro-aeration technology. Until now, these mechanisms have been speculated about in several studies, and could be summarized as the low solubility of oxygen in water, the rapid consumption of dissolved oxygen by facultative anaerobes, the formation of microbial aggregates, and the intrinsic tolerance of some methanogenic species to oxygen exposure (Figure 2).

First, oxygen has low solubility in water: only about 210 μM at 37°C in laboratory culture media [11]. Although high oxygen loads during micro-aeration might be applied, dissolved oxygen levels in the liquid phase are low (Figure 2A). Therefore, the oxygen content in the liquid phase that could influence the obligate methanogens remains rather low. Second, the AD system comprises various types of microbes that have unique functions with differing levels of tolerance to oxygen, which might be able to further reduce the oxygen levels in the liquid phase [58]. Facultative anaerobes are an important part of acidogenic bacterial communities in AD systems [59] and could consume the dissolved oxygen rapidly and result in a low redox potential that is necessary for obligate methanogens (Figure 2B) [19]. Third, the microorganisms in the AD system usually form microbial aggregates (e.g., flocs, granules, biofilms), which comprise the methanogens' dominant core and a bacteria-dominant surface (Figure 2C) [60]. Therefore, the methanogens would be well protected from oxygen stress due to the oxygen diffusion limitation and rapid oxygen consumption by the facultative or aerobic organisms thriving closer to the surface. Fourth, although oxygen was considered lethal for methanogens, some methanogens (e.g., the genera *Methanosarcina* and *Methanocella*) have shown the ability to survive oxygen stress [61,62]. As reported, methanogens could encode various antioxidant enzymes [e.g., superoxide reductase (SOR), SOD, rubredoxin oxidoreductase] and catalase for ROS (i.e., $O_2^-$, $H_2O_2$, and •OH) detoxification (Figure 2D).

Considering the indispensable role of methanogenesis in the AD process, it is essential to maintain the normal function of methanogens. Analysis of the mechanisms of how methanogens

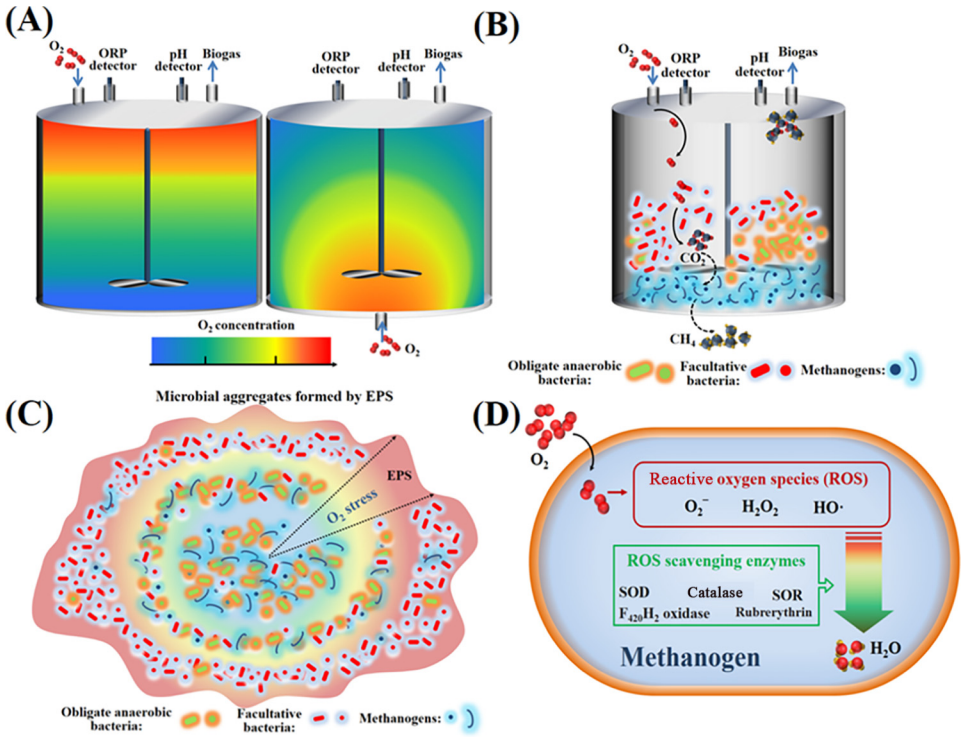



Figure 2. Survival mechanisms of methanogens under oxygen stress during micro-aeration. (A) Dissolved oxygen levels in the liquid phase are low due to low oxygen solubility in water. (B) Rapid consumption of dissolved oxygen by facultative anaerobes results in a low redox potential (C) Formation of microbial aggregates protects methanogens from oxygen stress due to the oxygen diffusion limitation and rapid oxygen consumption by the facultative or aerobic organisms thriving closer to the surface. (D) Intrinsic tolerance of some methanogens to oxygen stress. Abbreviations: EPS, extracellular polymeric substances; $F_{420}$, coenzyme $F_{420}$; ORP, oxidation-reduction potential; ROS, reactive oxygen species; SOD, superoxide dismutase; SOR, superoxide reductase.

survive under oxygen stress emphasizes that the inoculum used is crucial. An inoculum with a high abundance of facultative anaerobes (e.g., microorganisms from the phylum Firmicutes) could help in rapid consumption of the oxygen molecules, thus protecting the methanogens from oxygen stress and the secretion of additional extracellular hydrolases, thus benefiting substrate hydrolysis [63]. In addition to inoculum selection, the long-term domestication of the inoculum using micro-aeration could also cultivate a unique microbial community, which is able to function efficiently under micro-aeration conditions. The formation of microbial aggregates also favors the protection of methanogens from oxygen stress. Therefore, an inoculum rich in microbial aggregates (e.g., anaerobic granular sludge) could also be a potential choice for micro-aeration.

## Micro-aeration in AD: what are the next steps?

Micro-aeration is considered an easy and economical technology to facilitate AD due to the fact that it can be performed directly in the AD reactor with only limited oxygen or air needed. However, studies on the economic feasibility of micro-aeration technology are lacking, although micro-aeration is increasingly applied as a means to overcome specific challenges of AD such as increasing the biodegradability of lignocellulosic matter or the removal of specific contaminants. Until now, most of the studies on micro-aeration have been limited to the laboratory scale. Only a few studies have investigated the application performance of micro-aeration at full scale and

were mainly focused on the bioremoval of $H_2S$ in biogas [64,65]. Besides, several barriers need to be addressed to push micro-aeration technology to industrial application.

### Exploring efficient methods for oxygen dosing

Exogenous oxygen addition is the most-used micro-aeration method in the AD process. However, the low solubility of oxygen in water combined with the poor transfer of the gases to the liquid phases would pose serious barriers to this oxygen-dosing method. The dissolved oxygen represents a limiting factor for the growth of facultative anaerobes, as they can use only dissolved oxygen for extracellular hydrolase secretion. As Sheets and colleagues reported, limited air exposure to the solid-state AD reactor headspace did not significantly influence the methane yield from switchgrass due to oxygen diffusion limitation at the gas–liquid interphase [66]. In addition, a considerable portion of added oxygen would escape from the liquid phase, which could form dangerous and explosive gas mixtures if the methane content reached to between its **lower explosive limit (LEL)** (4.6%) and **upper explosive limit (UEL)** (15.8%) [67]. The residual oxygen in biogas could also be a barrier to the biogas upgrading process. Therefore, exploration of efficient methods for oxygen dosing is crucial for the development of micro-aeration technology.

Using oxygen-transferring membranes for controlled oxygen dosing is a potential option to solve the oxygen diffusion limitation problem at the gas–liquid interphase [68]. However, during micro-aeration these membranes succumb to membrane pore blocking and fouling problems. Studies on new material selection and surface modification should be conducted to move this technology closer to applications at scale. Using nanobubble aeration could be another solution to the oxygen diffusion limitation problem [69]. Nanobubbles are essentially gaseous spherical bubbles with diameters less than 1 μm, typically ranging between 50 and 200 nm, which have residence times of days to months in solution [69]. The unique characteristics of nanobubbles enable effective gas transfer and precise air/$O_2$ dosing control for micro-aeration technology [70]. Although the use of nanobubbles to create micro-aeration conditions has been investigated in several studies [70–72], mass transfer and the effects of nanobubbles on the microbial communities, microbial metabolism, enzyme activities, etc. have not been fully studied. Recent studies on the application of nanobubble technology are limited to batch and laboratory scale, while continuous and scale-up experiments should be developed to provide a reference for industrial-scale applications. Oxygen-releasing materials including oxygen-carrying materials (OCBs) and oxygen-generating materials (OGBs) could also be used for in-site oxygen supply [73]. For instance, our group recently used granular calcium peroxide ($CaO_2$) as an additive for in-site oxygen supply to create micro-aeration conditions and obtained an 11%-higher methane yield from corn straw [74]. However, the oxygen-releasing characteristics of oxygen-releasing materials and the interaction between the materials and the biochemical process associated with AD are largely unknown and need to be further investigated. Moreover, a new concept, named **electrolytic aeration** according to water electrolysis, provides a cutting-edge solution for in-site micro-aeration [10]. During electrolytic aeration, oxygen and hydrogen would be produced simultaneously, of which oxygen could be used for micro-aeration formation and hydrogen could be used as a substrate for methane production [75]. Meanwhile, the level of aeration could be precisely controlled by the electric current. This technique shows great potential for the enhancement of AD performance according to the micro-aeration concept as well the as realization of renewable surplus electricity storage according to the power-to-gas (P2G) concept [76], which must be investigated further to gain greater insight into its applicability.

### Revealing the mechanisms of how methanogens survive against oxygen stress

Methanogenesis is the final step of the AD process and is performed by methanogens. Methanogens are strictly anaerobes, which have been assumed to fail to protect themselves

from ROS such as superoxide and hydrogen peroxide [11]. However, studies have proved that some orders of methanogens (e.g., Methanocellales, Methanomicrobiales, Methanosarcinales) show intrinsic tolerance to oxidants [61,77]. Understanding the mechanisms of how methanogens survive oxygen stress could greatly promote our knowledge of AD theories. Several studies have tried to uncover the underlying mechanisms and obtained some interesting results. As reported, the oxygen-tolerant methanogens could encode various antioxidant enzymes (e.g., SOR, SOD, rubredoxin oxidoreductase) and catalase for ROS (i.e., $O_2^-$, $H_2O_2$, and •OH) detoxification [61,62], thus protecting themselves from oxygen stress. However, studies on the mechanisms of how methanogens survive oxygen stress remain in their infancy. The reactions and responses of different methanogen species to oxygen stress and the effects of long-term oxygen exposure on the physiological and biochemical characteristics of different methanogen species, as well as the possibility of long-term domestication of methanogens to micro-aeration condition are still poorly reported. In addition, oxygen flow direction during micro-aeration is not fully revealed in previous studies. Although the oxygen was speculated to be used by the facultative anaerobes for their metabolism, no direct evidence is available to prove this speculation. Using state-of-the-art techniques such as genome-centric metagenomics and metatranscriptomics, stable isotope labeling to analyze the mechanisms of how methanogens survive oxygen stress as well as the oxygen flow direction during micro-aeration would be helpful in gaining deeper insight into micro-aeration technology and guiding its implementation.

### Long-term effects of micro-aeration on AD and pilot or industrial-scale experiments

Using micro-aeration to regulate the AD of different organic wastes (e.g., cellulosic substrate, sewage sludge, wastewater, black water) has been widely reported, but most of the studies were conducted at batch or laboratory scale. AD is a biochemical process performed by various microorganisms, which are easily altered by operational parameters [59]. During the long-term application of micro-aeration, the microorganisms of the AD would be screened and domesticated to form a unique microbial community. Thus, the long-term effects of micro-aeration on AD and microbial community succession under micro-aeration stress should be investigated. Furthermore, hydromechanical properties, mass transfer, and biochemical reactions could differ between laboratory, pilot, or industrial-scale experiments. Thus, pilot or industrial-scale experiments should also be conducted to promote the industrial application of the micro-aeration technology.

## Concluding remarks

Micro-aeration, the introduction of limited oxygen/air into an AD system, represents a promising strategy to improve the AD process. Micro-aeration has shown great potential to enhance hydrolysis, improve methane yield, stabilize AD, and strip hydrogen sulfide. Compared with other techniques (e.g., substrate pretreatment, bioaugmentation, specific enzyme addition), micro-aeration is expected to be more environmentally friendly and economical since only limited oxygen or air is needed and it can be performed on-site in the AD reactor. Our group recently conducted a preliminary economic analysis of different pretreatment methods including alkali hydrogen peroxide pretreatment, NaOH pretreatment, HCl pretreatment, hot water pretreatment, and micro-aerobic pretreatment, and found that only micro-aerobic pretreatment was economically feasible compared with an un-pretreated group [78]. However, comprehensive economic assessment of micro-aeration technology remains lacking. Along with the increasing expansion of the biogas industry, micro-aeration technology requires rapid development.

Despite recent advances, several challenges need to be addressed and certain questions need to be answered (see Outstanding questions) to move micro-aeration technology closer to practical application. Scaling up and long-term running are urgently needed as hydrodynamics and mass transport will vary greatly during the scaling up process. Some approaches [e.g., computational

### Outstanding questions

What is the biggest barrier that limits the industrial application of micro-aeration technology? How can we move micro-aeration technology from the laboratory scale to industrial application?

Should additional steps be added to current mathematical models to simulate outcomes of micro-aeration?

How do different microorganisms cope with oxygen stress? Do syntrophic relationships exist under micro-aeration conditions?

How can we control micro-aeration intensity (e.g., oxygen dosage, oxygen exposure pathway)? Do we have a simple and easy method to control micro-aeration intensity?

fluid dynamics (CFD) methods] could be adopted to assist micro-aeration reactor design and oxygen-dosing optimization. Another concern is associated with the biological understanding of micro-aeration. Micro-aeration is speculated to mainly work in microbial community regulation, which includes a series of biochemical reactions including enzyme activity variations, syntrophic collaboration establishment between different microorganisms, etc. Although some studies have reported the effects of micro-aeration on microbial communities, enzyme activity, intermediate products, etc., we still know little about the process. Advanced techniques such as genome-centric metatranscriptomics and stable isotope labeling could provide us with a wealth of information about what happens during micro-aeration and guide the application of micro-aeration technology in turn. In addition, other beneficial effects of micro-aeration on AD – for example antimicrobial resistance gene (ARG) control [79], alleviation of ammonia inhibition [80], etc. – have been reported recently. Thus, studies to comprehensively analyze the beneficial effects of micro-aeration on AD are also needed to deepen our understanding.

In light of the great potential of micro-aeration in facilitating AD, we trust that practical application will come in the near future due to the increased research and development of advanced techniques that warrant bringing this technology to the next level.

## Acknowledgments

The project was supported by the National Natural Science Foundation of China (51808260), the State Key Laboratory of Pollution Control and Resource Reuse Foundation (No. PCRRF21044), 'Transformational Technologies for Clean Energy and Demonstration', Strategic Priority Research Program of the Chinese Academy of Sciences (XDA 21060400), and the Natural Science Foundation of Shandong Province (ZR2020ME233; ZR2020ME234; and ZR2020KE043).

## Declaration of interests

No interests are declared.

## References

1. Náthia-Neves, G. *et al.* (2018) Anaerobic digestion process: technological aspects and recent developments. *Int. J. Environ. Sci. Technol.* 15, 2033–2046
2. Merlin, G. and Boileau, H. (2013) Anaerobic digestion of agricultural waste: state of the art and future trends. In *Anaerobic digestion: types, processes and environmental impact* (Torrles, A., ed.), Nova Science
3. Zhang, T. *et al.* (2022) New insight on Fe-bioavailability: bio-uptake, utilization and induce in optimizing methane production in anaerobic digestion. *Chem. Eng. J.* 441, 136099
4. Xiao, L. *et al.* (2022) Enhanced methane production by granular activated carbon: a review. *Fuel* 320, 123903
5. Abanades, S. *et al.* (2022) A critical review of biogas production and usage with legislations framework across the globe. *Int. J. Environ. Sci. Technol. (Tehran)* 19, 3377–3400
6. Fu, S. *et al.* (2021) *In situ* biogas upgrading by $CO_2$-to-$CH_4$ bioconversion. *Trends Biotechnol.* 39, 336–347
7. Nguyen, D. *et al.* (2019) Intermittent micro-aeration: new strategy to control volatile fatty acid accumulation in high organic loading anaerobic digestion. *Water Res.* 166, 115080
8. Verstraete, W. *et al.* (2007) Microbial resource management: the road to go for environmental biotechnology. *Eng. Life Sci.* 7, 117–126
9. Carballa, M. *et al.* (2015) Microbial management of anaerobic digestion: exploiting the microbiome–functionality nexus. *Curr. Opin. Biotechnol.* 33, 103–111
10. Chen, Q. *et al.* (2020) Review on microaeration-based anaerobic digestion: state of the art, challenges, and prospectives. *Sci. Total Environ.* 710, 136388
11. Lu, Z. and Imlay, J.A. (2021) When anaerobes encounter oxygen: mechanisms of oxygen toxicity, tolerance and defence. *Nat. Rev. Microbiol.* 19, 774–785
12. Xu, H. *et al.* (2021) Effect of microaerobic microbial pretreatment on anaerobic digestion of a lignocellulosic substrate under controlled pH conditions. *Bioresour. Technol.* 328, 124852
13. Zeb, I. *et al.* (2022) *In-situ* microaeration of anaerobic digester treating buffalo manure for enhanced biogas yield. *Renew. Energy* 181, 843–850
14. Andreides, M. *et al.* (2021) Stirring-based control strategy for microaerobic $H_2S$ removal in sequencing batch anaerobic digesters. *Fuel* 306, 121696
15. Diaz, I. *et al.* (2011) Effect of microaerobic conditions on the degradation kinetics of cellulose. *Bioresour. Technol.* 102, 10139–10142
16. Fdz-Polanco, M. *et al.* (2009) Hydrogen sulphide removal in the anaerobic digestion of sludge by micro-aerobic processes: pilot plant experience. *Water Sci. Technol.* 60, 3045–3050
17. Li, J. *et al.* (2022) Oxidative stress and antioxidant mechanisms of obligate anaerobes involved in biological waste treatment processes: a review. *Sci. Total Environ.* 838, 156454
18. Khademian, M. and Imlay, J.A. (2021) How microbes evolved to tolerate oxygen. *Trends Microbiol.* 29, 428–440
19. Nguyen, D. and Khanal, S.K. (2018) A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. *Biotechnol. Adv.* 36, 1971–1983
20. Liu, C.G. *et al.* (2013) Redox potential control and applications in microaerobic and anaerobic fermentations. *Biotechnol. Adv.* 31, 257–265
21. Kato, M.T. *et al.* (1993) Methanogenesis in granular sludge exposed to oxygen. *FEMS Microbiol. Lett.* 114, 317–324
22. Shen, C.F. and Guiot, S.R. (1996) Long-term impact of dissolved $O_2$ on the activity of anaerobic granules. *Biotechnol. Bioeng.* 49, 611–620
23. Shen, C.F. *et al.* (1996) Methanotroph and methanogen coupling in granular biofilm under $O_2$-limited conditions. *Biotechnol. Lett.* 18, 495–500
24. Cheng, F. and Brewer, C.E. (2021) Conversion of protein-rich lignocellulosic wastes to bio-energy: review and recommendations for hydrolysis + fermentation and anaerobic digestion. *Renew. Sust. Energ. Rev.* 146, 111167

25. Ariunbaatar, J. *et al.* (2014) Pretreatment methods to enhance anaerobic digestion of organic solid waste. *Appl. Energy* 123, 143–156
26. Fu, S.-F. *et al.* (2016) Impacts of microaeration on the anaerobic digestion of corn straw and the microbial community structure. *Chem. Eng. J.* 287, 523–528
27. Duarte, M.S. *et al.* (2018) Insight into the role of facultative bacteria stimulated by microaeration in continuous bioreactors converting LCFA to methane. *Environ. Sci. Technol.* 52, 6497–6507
28. Hanreich, A. *et al.* (2013) Metagenome and metaproteome analyses of microbial communities in mesophilic biogas-producing anaerobic batch fermentations indicate concerted plant carbohydrate degradation. *Syst. Appl. Microbiol.* 36, 330–338
29. Ariesyady, H.D. *et al.* (2007) Functional bacterial and archaeal community structures of major trophic groups in a full-scale anaerobic sludge digester. *Water Res.* 41, 1554–1568
30. Zhu, R. *et al.* (2022) Understanding the mechanisms behind micro-aeration to enhance anaerobic digestion of corn straw. *Fuel* 318, 3604
31. Lim, J.W. *et al.* (2014) Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste. *Bioresour. Technol.* 171, 132–138
32. Xu, S. *et al.* (2014) Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. *Waste Manag.* 34, 363–369
33. Zhu, R. *et al.* (2022) The effects of micro-aeration on semi-continued anaerobic digestion of corn straw with increasing organic loading rates. *Renew. Energy* 195, 1194–1201
34. Yang, H. and Deng, L. (2020) Using air instead of biogas for mixing and its effect on anaerobic digestion of animal wastewater with high suspended solids. *Bioresour. Technol.* 318, 124047
35. Wu, Z. *et al.* (2021) Synergistic association between cytochrome *bd*-encoded *Proteiniphilum* and reactive oxygen species (ROS)-scavenging methanogens in microaerobic–anaerobic digestion of lignocellulosic biomass. *Water Res.* 190, 116721
36. Ruan, R. *et al.* (2017) Performance evaluation of combining microaerobic desulphurization with addition of rusty scrap iron (RSI) during waste-activated sludge digestion. *Desalin. Water Treat.* 84, 169–179
37. Krayzelova, L. *et al.* (2015) Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. *Rev. Environ. Sci. Biotechnol.* 14, 703–725
38. Heidelberg, J.F. *et al.* (2004) The genome sequence of the anaerobic, sulfate-reducing bacterium *Desulfovibrio vulgaris* Hildenborough. *Nat. Biotechnol.* 22, 554–559
39. Hirasawa, J.S. *et al.* (2008) Application of molecular techniques to evaluate the methanogenic archaea and anaerobic bacteria in the presence of oxygen with different COD:sulfate ratios in a UASB reactor. *Anaerobe* 14, 209–218
40. Muyzer, G. and Stams, A.J.M. (2008) The ecology and biotechnology of sulphate-reducing bacteria. *Nat. Rev. Microbiol.* 6, 441–454
41. Huertas, J.K. *et al.* (2020) Comparing hydrogen sulfide removal efficiency in a field-scale digester using microaeration and iron filters. *Energies* 13, 4793
42. Ramos, I. *et al.* (2012) The role of the headspace in hydrogen sulfide removal during microaerobic digestion of sludge. *Water Sci. Technol.* 66, 2258–2264
43. Ramos, I. *et al.* (2014) The headspace of microaerobic reactors: sulphide-oxidising population and the impact of cleaning on the efficiency of biogas desulphurisation. *Bioresour. Technol.* 158, 63–73
44. Giordano, A. *et al.* (2019) Long-term biogas desulfurization under different microaerobic conditions in full-scale thermophilic digesters co-digesting high-solid sewage sludge. *Int. Biodeterior. Biodegradation* 142, 131–136
45. Castro, I.M.P. *et al.* (2021) Evaluation of microaeration strategies in the digestion zone of UASB reactors as an alternative for biogas desulfurization. *Environ. Technol.* Published online September 10, 2021. https://doi.org/10.1080/09593330.2021.1974574
46. Mahdy, A. *et al.* (2020) Simultaneous $H_2S$ mitigation and methanization enhancement of chicken manure through the introduction of the micro-aeration approach. *Chemosphere* 253, 126687
47. Song, Y. *et al.* (2020) Air supplement as a stimulation approach for the *in situ* desulfurization and methanization enhancement of anaerobic digestion of chicken manure. *Energy Fuel* 34, 12606–12615
48. Cheng, Z. *et al.* (2018) Enhancement of surfactant biodegradation with an anaerobic membrane bioreactor by introducing microaeration. *Chemosphere* 208, 343–351
49. Siqueira, J.P.S. *et al.* (2018) Process bioengineering applied to BTEX degradation in microaerobic treatment systems. *J. Environ. Manag.* 223, 426–432
50. Zhang, Z. *et al.* (2022) Coupling of membrane-based bubbleless micro-aeration for 2,4-dinitrophenol degradation in a hydrolysis acidification reactor. *Water Res.* 212, 118119
51. Zhang, C. and Bennett, G.N. (2005) Biodegradation of xenobiotics by anaerobic bacteria. *Appl. Microbiol. Biotechnol.* 67, 600–618
52. Lian, S. *et al.* (2021) Accelerated adsorption of tetracyclines and microbes with $FeO_n(OH)_m$ modified oyster shell: its application on biotransformation of oxytetracycline in anaerobic enrichment culture. *Chem. Eng. J.* 425, 130499
53. Raymond, J. and Segrè, D. (2006) The effect of oxygen on biochemical networks and the evolution of complex life. *Science* 311, 1764–1767
54. Meunier, B. *et al.* (2004) Mechanism of oxidation reactions catalyzed by cytochrome P450 enzymes. *Chem. Rev.* 104, 3947–3980
55. Yeom, S.J. *et al.* (2022) P450-driven plastic-degrading synthetic bacteria. *Trends Biotechnol.* 40, 166–179
56. Jia, Y. *et al.* (2018) Ciprofloxacin degradation in anaerobic sulfate-reducing bacteria (SRB) sludge system: mechanism and pathways. *Water Res.* 136, 64–74
57. Behrendorff, J. (2021) Reductive cytochrome P450 reactions and their potential role in bioremediation. *Front. Microbiol.* 12, 649273
58. Achinas, S. *et al.* (2020) Microbiology and biochemistry of anaerobic digesters: an overview. *Bioreactors* 2020, 17–26
59. Laiq Ur Rehman, M. *et al.* (2019) Anaerobic digestion. *Water Environ. Res.* 91, 1253–1271
60. Sekiguchi, Y. *et al.* (1999) Fluorescence *in situ* hybridization using 16S rRNA-targeted oligonucleotides reveals localization of methanogens and selected uncultured bacteria in mesophilic and thermophilic sludge granules. *Appl. Environ. Microbiol.* 65, 1280–1288
61. Lyu, Z. and Lu, Y. (2018) Metabolic shift at the class level sheds light on adaptation of methanogens to oxidative environments. *ISME J.* 12, 411–423
62. Andrade, K. *et al.* (2015) Metagenomic and lipid analyses reveal a diel cycle in a hypersaline microbial ecosystem. *ISME J.* 9, 2697–2711
63. Fu, S.-F. *et al.* (2016) Effect of different mixed microflora on the performance of thermophilic microaerobic pretreatment. *Energy Fuel* 30, 6413–6418
64. Jenicek, P. *et al.* (2017) Simple biogas desulfurization by microaeration – full scale experience. *Anaerobe* 46, 41–45
65. Jenicek, P. *et al.* (2010) Advantages of anaerobic digestion of sludge in microaerobic conditions. *Water Sci. Technol.* 62, 427–434
66. Sheets, J.P. *et al.* (2015) Effect of limited air exposure and comparative performance between thermophilic and mesophilic solid-state anaerobic digestion of switchgrass. *Bioresour. Technol.* 180, 296–303
67. Kundu, S. *et al.* (2016) A review on understanding explosions from methane–air mixture. *J. Loss Prev. Process Ind.* 40, 507–523
68. Werkneh, A.A. (2022) Application of membrane-aerated biofilm reactor in removing water and wastewater pollutants: current advances, knowledge gaps and research needs – a review. *Environ. Chall.* 8, 100529
69. Chuenchart, W. *et al.* (2021) Nanobubble technology in anaerobic digestion: a review. *Bioresour. Technol.* 329, 124916
70. Wang, X. *et al.* (2020) Improved methane production from corn straw using anaerobically digested sludge pre-augmented by nanobubble water. *Bioresour. Technol.* 311, 123479

71. Wang, X. *et al.* (2020) Supplementation of $O_2$-containing gas nanobubble water to enhance methane production from anaerobic digestion of cellulose. *Chem. Eng. J.* 398, 125652
72. Xiao, W. and Xu, G. (2020) Mass transfer of nanobubble aeration and its effect on biofilm growth: microbial activity and structural properties. *Sci. Total Environ.* 703, 134976
73. Willemen, N.G.A. *et al.* (2021) Oxygen-releasing biomaterials: current challenges and future applications. *Trends Biotechnol.* 39, 1144–1159
74. Fu, S.-F. *et al.* (2018) Improved methane production of corn straw by the stimulation of calcium peroxide. *Energy Convers. Manag.* 164, 36–41
75. Tartakovsky, B. *et al.* (2011) Electrolysis-enhanced anaerobic digestion of wastewater. *Bioresour. Technol.* 102, 5685–5691
76. Geppert, F. *et al.* (2016) Bioelectrochemical power-to-gas: state of the art and future perspectives. *Trends Biotechnol.* 34, 879–894
77. Horne, A.J. and Lessner, D.J. (2013) Assessment of the oxidant tolerance of *Methanosarcina acetivorans*. *FEMS Microbiol. Lett.* 343, 13–19
78. Fu, S.-F. *et al.* (2018) Improved anaerobic digestion performance of *Miscanthus floridulus* by different pretreatment methods and preliminary economic analysis. *Energy Convers. Manag.* 159, 121–128
79. Guo, B. *et al.* (2021) Effects of micro-aeration on microbial niches and antimicrobial resistances in blackwater anaerobic digesters. *Water Res.* 196, 117035
80. Fan, Y. *et al.* (2021) Alleviation of ammonia inhibition via nanobubble water supplementation during anaerobic digestion of ammonia-rich swine manure: buffering capacity promotion and methane production enhancement. *Bioresour. Technol.* 333, 125131